

Dead reckoning. This

is an essential navigation technique for estimating the current position of a moving object based on its previous known position (last fix), movement data (speed, heading, and time), which are commonly employed when GPS data is unavailable. It is also known as “deduced reckoning” or “DR.”

Installing Hector SLAM

Before installing the Hector SLAM, first, install the robot operating system (ROS).

Check at <http://surl.li/jqmux> to learn how to do so.

Subsequently, set up the drivers and software development kit (SDK) for your lidar. If utilising the RPI lidar, you can directly proceed to Hector SLAM installation. However, if employing the YD lidar, continue with the following steps:

```
git clone https://github.com/YDLIDAR/YDLidar-SDK
cd YDLidar-SDK/build
cmake .
Make sudo make install
```

Next, create the ROS workspace for the lidar and then install the ROS driver using the following commands in the LXTerminal:

```
mkdir -p ~/ydlidar_ws/src
cd ~/ydlidar_ws
catkin_make
echo "source ~/ydlidar_ws/devel/setup."
bash" >> ~/.bashrc
source ~/.bashrc
cd ~/ydlidar_ws/src/ydlidar_ros_driver/
startup
```

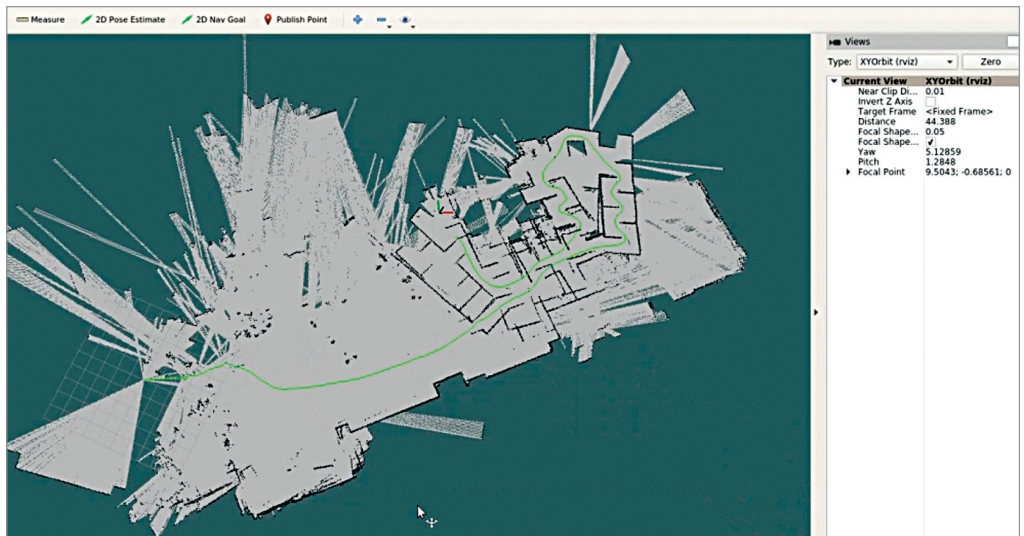


Fig. 5: ROS Hector SLAM showing maps path captured using lidar

```
sudo chmod +x initenv.sh
sudo sh initenv.sh
```

Integrating Hector SLAM

Open a new LXTerminal and run the following command to install Hector SLAM:

```
sudo apt-get install ros-kinetic-hector-
slam

wgethttps://storage.googleapis.com/
google-code-archive-downloads/v2/code.
google.com/tu-darmstadt-ros-pkg/Team_
Hector_MappingBox_RoboCup_2011_Rescue_
Arena.bag
```

The Hector SLAM is now ready. Modify some lidar settings in the lidar launch file for indoor positioning. To do so, first, go to the YD lidar workspace and open the src file. Then open all nodes.launch file and change the parameters as per Fig. 3, which shows configuration of the ROS nodes according to lidar.

Testing

Connect the YD lidar to the USB port of the RPi board/Nvidia Jetson

board, power it, and access its desktop GUI. For positioning, execute the following command in the LXTerminal:

```
roslaunch ydlidar_ros_driver all_nodes.  
launch
```

This command will generate a map showcasing the robot's current position and its immediate surroundings. Fig. 4 presents the SLAM data, while Fig. 5 illustrates the ROS Hector SLAM's depiction of mapped paths captured via lidar.

In next segment, we will merge this ROS environment with the previous lane detection system. By doing so, we will establish an integrated framework for implementing autonomous driving and the ADAS system.

To test the Hector SLAM and obtain a real-time map, run the following commands:

```
roslaunch hector_slam_launch tutorial.  
launch  
rosbag play Team_Hector_MappingBox_  
RoboCup_2011_Rescue_Arena.bag --clock  
rostopic pub syscommand std_msgs/String  
"savegeotiff" EFY
```

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EFY Note

The source code of this project is available for free download at www.electronicsforu.com